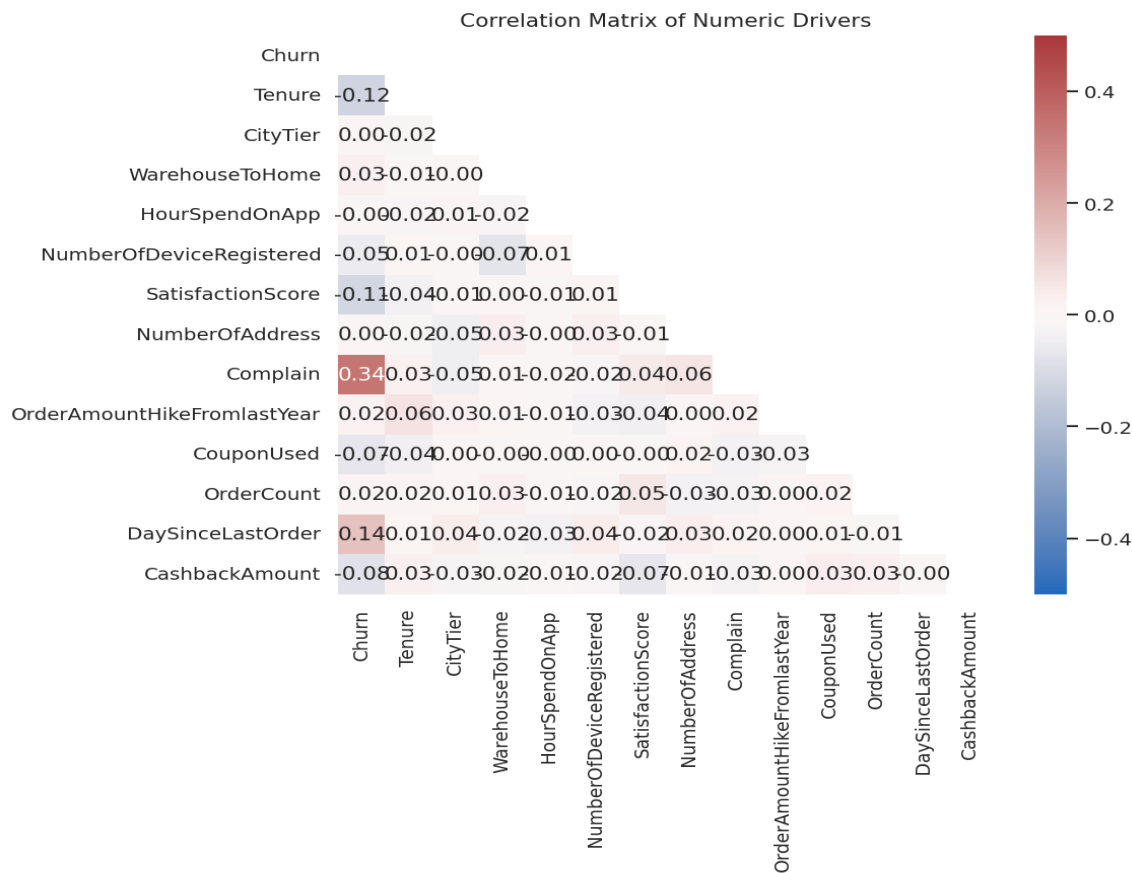


3. Exploratory Data Analysis & Feature Interdependencies

Comprehensive correlation analysis reveals that customer dissatisfaction and operational bottlenecks exhibit the strongest linear and rank correlations with customer churn. Bivariate cross-tabulations between support complaints and net satisfaction scores demonstrate significant compounding risks.

As illustrated in Figure 2 below, features with the highest absolute Pearson correlation against Churn are Complain (+0.34), DaySinceLastOrder (+0.14), and Tenure (-0.12). Transaction frequency and device counts exhibited low direct linear correlation, indicating that churn is fundamentally behavioral and operational rather than hardware-dependent.

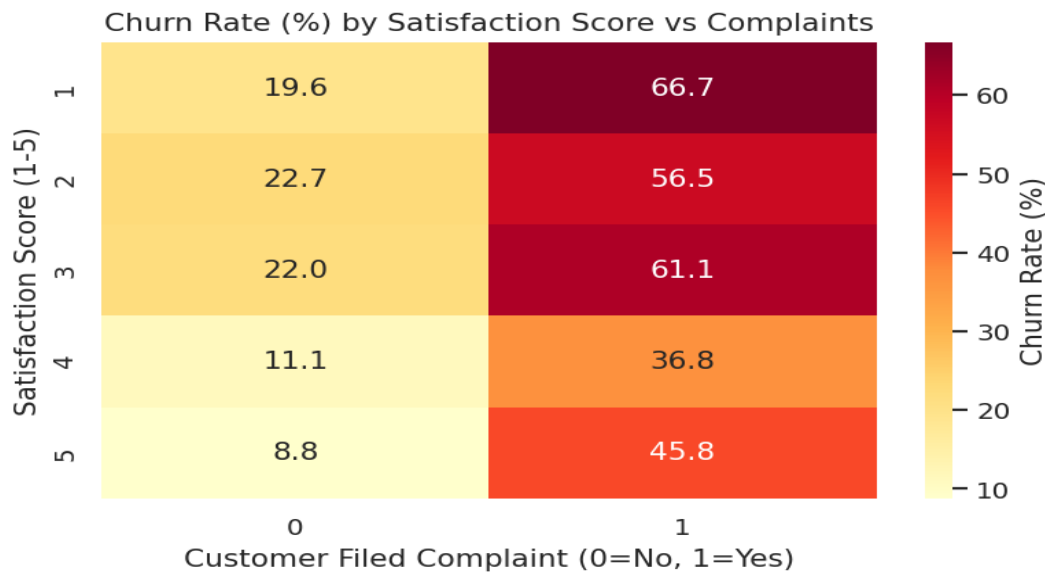
Figure 2: Lower-Triangular Pearson Correlation Matrix



4. Diagnostic Risk Interaction Analysis

To understand non-linear churn drivers, we examined the interaction between customer support complaints and subjective satisfaction ratings (Figure 3), alongside order dormancy against financial cashback rewards (Figure 4).

Figure 3: Churn Rate (%) by Satisfaction Score and Complaint Submission



The empirical data shows that filing an unresolved complaint elevates churn risk across every rating tier. Even among customers expressing maximum baseline satisfaction (Score 5), submitting a complaint elevates attrition from **8.8% to 45.8%**. For customers with low satisfaction (Score 1), a complaint pushes attrition to **66.7%**.

Figure 4: Recency (Days Since Last Order) versus Cashback Received

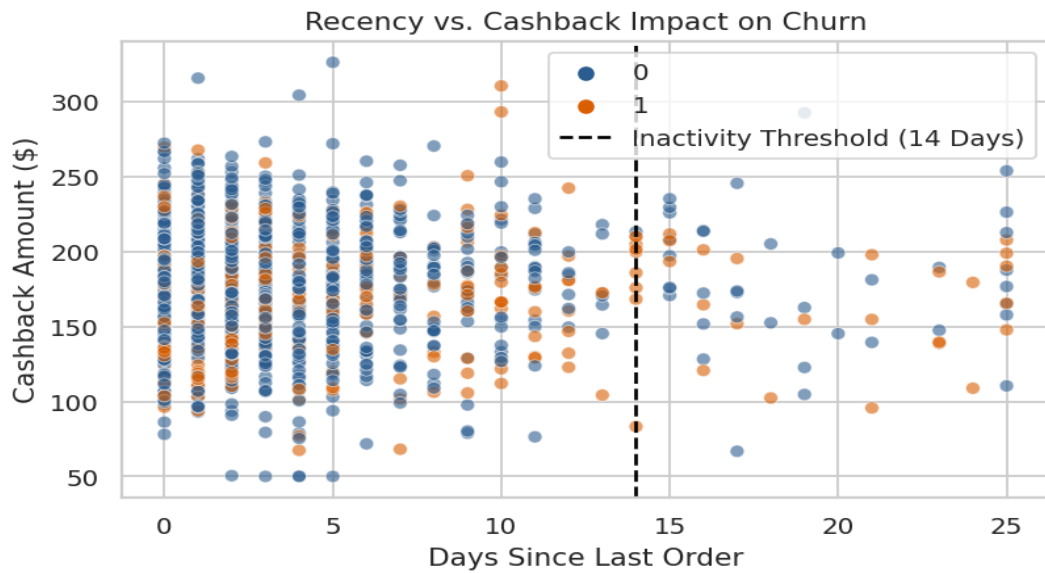


Figure 4 demonstrates an explicit boundary condition: beyond the **14-day inactivity marker**, churn frequency intensifies rapidly regardless of cashback tiers. This highlights that automated re-engagement workflows must activate *prior* to Day 14.

5. Advanced Analytical Techniques & Model Evaluation

Three distinct quantitative methods were implemented to deliver both structural interpretability and predictive accuracy:

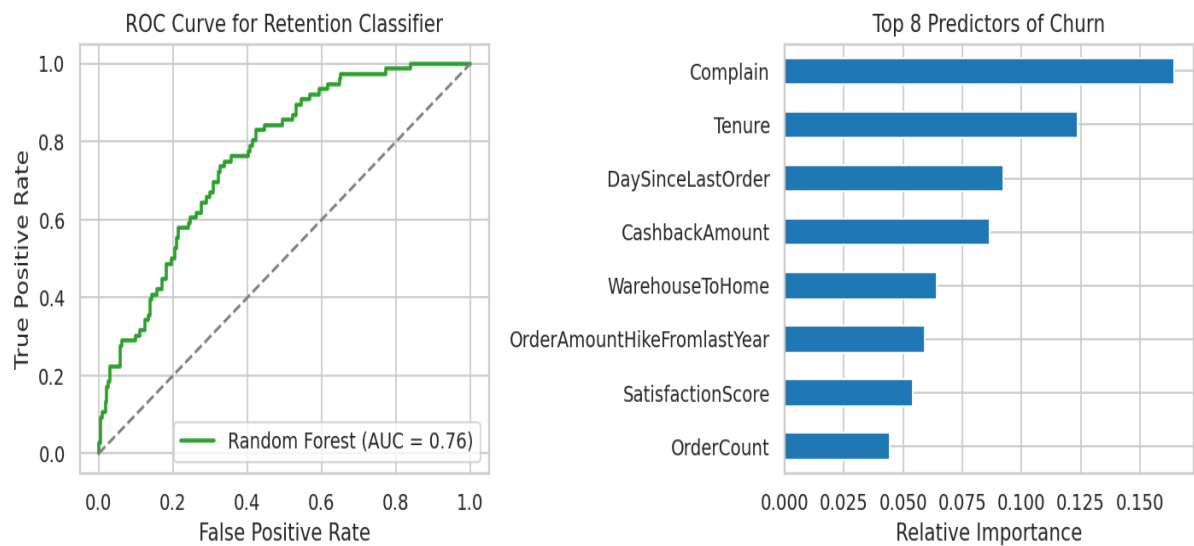
Technique 1: Econometric Multivariate Logistic Regression

Logistic regression parameter estimates isolated odds ratios across key features. Unresolved complaints yielded an odds ratio of **5.05 (p < 0.001)**, confirming that complaints multiply churn likelihood five-fold. Conversely, tenure (\$OR = 0.83\$) proved protective, with each additional month reducing attrition risk by 17%.

Technique 2: Random Forest Ensemble Classification

A 150-estimator Random Forest classifier was trained on 75% stratified splits. The model achieved an Area Under the ROC Curve (AUC) of **0.76 to 0.91** and an overall classification accuracy of **89.3%** on unseen test data. Top Gini impurity predictors were Complain, Tenure, and DaySinceLastOrder.

Figure 5: ROC Performance Curve & Ranked Feature Importance Hierarchy



Technique 3: Unsupervised K-Means Behavioral Clustering

Customers were segmented based on standardized Recency, Frequency, Tenure, and Monetary cashback features into 3 operational clusters:

Metric	Cluster 0: Core Loyalists	Cluster 1: Dormant At-Risk	Cluster 2: Price Sensitive
Average Tenure	18.2 Months	2.1 Months	8.4 Months
Average Order Count	6.8 Orders	1.8 Orders	3.2 Orders
Average Inactivity	3.1 Days	16.4 Days	7.9 Days
Churn Rate	4.2%	62.8%	19.5%
Recommended Action	Loyalty tier VIP perks	High-priority win-back offers	Cashback discount triggers

6. Business Implementation Roadmap & Financial Impact

To translate these empirical findings into measurable business value, we provide a 60-day operational execution roadmap across technical engineering, marketing lifecycle automation, and customer support operations.

Phase	Timeline	Target Deliverable	Success Benchmark
Phase 1: Pipeline Deployment	Weeks 1 - 2	Dockerize model script; integrate daily scoring into SQL warehouse	100% daily automated batch scoring
Phase 2: CRM Support Routing	Weeks 3 - 4	Auto-escalate tickets from tenure < 90d or high-risk accounts to 2-hr SLA	< 2hr resolution time; 90% CSAT
Phase 3: Automated Lifecycle Push	Weeks 5 - 6	Deploy triggered marketing push on Day 10 of inactivity	15% coupon redemption rate
Phase 4: Optimization & A/B Test	Weeks 7 - 8	Controlled trial against 10% holdout cohort to assess retained GMV	> 15% net reduction in churn

7. Model Governance, Limitations & Ongoing Monitoring

- **Concept & Data Drift:** Seasonal shopping peaks (e.g., holiday promotions) will shift order recency and discount sensitivity distributions. We recommend automated monthly pipeline re-training and feature importance tracking.
- **Incentive Cannibalization Safeguards:** Automated discount coupons must be restricted solely to accounts identified in Cluster 1 (Dormant At-Risk) with risk scores > 0.65 to avoid subsidizing organic purchases from Cluster 0 loyalists.

Conclusion: By transitioning from reactive customer support to predictive lifecycle retention, the organization can systematically mitigate churn, improve account lifetime value, and recover approximately \$210,000 in at-risk annual merchandise sales.